

Mathematics Curriculum — Master Index

Մաթեմատիկայի ուսումնական ծրագիր — ընդհանուր ցանկ

SCOPE	Grade 1 through university-level preparation · 22 courses
BANDS	Elementary · Middle School · High School · Advanced Placement · Examination Preparation · Competition
ALIGNMENT	American academic standards, with a Վարժարան progression designed for one-to-one instruction
DOCUMENT	This index maps the full catalog. Each course has its own complete syllabus.
REVISION	August 2026

HOW THIS CURRICULUM IS USED

Ինչպես է կիրառվում այս ծրագիրը

Every syllabus in this catalog defines the **standard course**. No student receives the standard course unchanged. After the placement assessment, Վարժարան produces a personalized version of the syllabus for that student: units they have already mastered are compressed or removed, gaps found upstream are inserted before the content that depends on them, and the pacing track is chosen from the options each syllabus lists. The personalized syllabus is visible to the parent in the parent portal and is revised whenever the progress profile changes materially.

This is the difference between Վարժարան and a tutoring marketplace. A tutor delivers lessons. A school delivers a curriculum, assesses against it, reports progress in terms of it, and adjusts it when the evidence says to. The syllabi in this catalog are what make that possible: they state, in advance and in public, what will be taught, what mastery looks like, how it will be measured, and what the student will be able to do at the end.

COURSE CATALOG

Դասընթացների ցանկ

CODE	COURSE	BAND	TYPICAL LEVEL	PREREQUISITE
MTH-G1	Grade 1 Mathematics · Մաթեմատիկա — 1-ին դասարան	Elem.	Grade 1	Counts to 20; numerals 0-10
MTH-G2	Grade 2 Mathematics · Մաթեմատիկա — 2-րդ դասարան	Elem.	Grade 2	MTH-G1
MTH-G3	Grade 3 Mathematics · Մաթեմատիկա — 3-րդ դասարան	Elem.	Grade 3	MTH-G2
MTH-G4	Grade 4 Mathematics · Մաթեմատիկա — 4-րդ դասարան	Elem.	Grade 4	MTH-G3

CODE	COURSE	BAND	TYPICAL LEVEL	PREREQUISITE
MTH-G5	Grade 5 Mathematics · Մաթեմատիկա — 5-րդ դասարան	Elem.	Grade 5	MTH-G4
MTH-G6	Grade 6 Mathematics · Մաթեմատիկա — 6-րդ դասարան	Middle	Grade 6	MTH-G5
MTH-G7	Grade 7 Mathematics · Մաթեմատիկա — 7-րդ դասարան	Middle	Grade 7	MTH-G6
MTH-G8	Grade 8 Mathematics · Մաթեմատիկա — 8-րդ դասարան	Middle	Grade 8	MTH-G7
MTH-PA	Pre-Algebra · Նախահանրահաշիվ	Middle	Grades 7-9 · bridge	MTH-G6 or placement
MTH-A1	Algebra I · Հանրահաշիվ I	High	Grades 8-9	MTH-G8 or MTH-PA
MTH-GE	Geometry · Երկրաչափություն	High	Grades 9-10	MTH-A1
MTH-A2	Algebra II · Հանրահաշիվ II	High	Grades 10-11	MTH-A1 + MTH-GE
MTH-TR	Trigonometry · Եռանկյունաչափություն	High	Grades 10-12 · standalone	MTH-A2 or concurrent
MTH-PC	Precalculus · Նախամաթանալիզ	High	Grades 11-12	MTH-A2
MTH-ST	Statistics · Վիճակագրություն	High	Grades 10-12 · standalone	MTH-A1
MTH-CA	Calculus · Մաթեմատիկական անալիզ	High	Grades 11-12	MTH-PC
MTH-APCA	AP Calculus AB · AP Մաթեմատիկական անալիզ AB	AP	Grades 11-12	MTH-PC
MTH-APCB	AP Calculus BC · AP Մաթեմատիկական անալիզ BC	AP	Grades 11-12	MTH-PC or MTH-APCA
MTH-APS	AP Statistics · AP Վիճակագրություն	AP	Grades 10-12	MTH-A2
MTH-SAT	SAT Mathematics · SAT մաթեմատիկա	Exam prep	Grades 10-12	MTH-A1 + MTH-GE
MTH-ACT	ACT Mathematics · ACT մաթեմատիկա	Exam prep	Grades 10-12	MTH-A1 + MTH-GE
MTH-COMP	Competition Mathematics · Օլիմպիադային մաթեմատիկա	Contest	Grades 5-12 · 3 levels	Placement only

PATHWAYS THROUGH THE CURRICULUM

Ուսումնական ուղիներ

The same catalog supports several routes. The pathway is assigned after the placement assessment and reviewed at each end-of-course assessment; a student may move between pathways in either direction without repeating content, because the courses themselves do not change — only the sequence and the pace do.

STANDARD

G1 → G2 → G3 → G4 → G5 → G6 → G7 → G8 → Algebra I → Geometry → Algebra II → Precalculus

Algebra I in Grade 9. Precalculus in Grade 12. Calculus is not reached in secondary school. This is the majority pathway in American public schools and is entirely sufficient for most university programmes.

ACCELERATED

... → G6 → G7 → G8 (compacted) → Algebra I (Gr 8) → Geometry → Algebra II → Precalculus → AP Calculus AB

Algebra I in Grade 8, reached by compacting Grades 7 and 8 across two years. Ends with AP Calculus AB in Grade 12. Recommended when the Grade 6 end-of-course assessment shows strong ratio reasoning and computational fluency.

ADVANCED

... → G5 → G6/G7 compacted → Algebra I (Gr 7) → Geometry → Algebra II → Precalculus → AP Calculus BC → university mathematics

Algebra I in Grade 7. Reaches AP Calculus BC in Grade 11 and leaves Grade 12 open for university-level coursework. Requires a strong Grade 5 profile and, in practice, genuine enthusiasm rather than only capability — this pathway is a poor fit for a capable student who does not want it.

STATISTICS-FOCUSED

... → Algebra I → Geometry → Algebra II → Statistics → AP Statistics

For students heading toward the social sciences, life sciences, business, medicine, or journalism, where statistical literacy matters more than calculus. AP Statistics requires only Algebra II and may be taken as early as Grade 10.

EXAMINATION

Any pathway + SAT Mathematics or ACT Mathematics in Grades 10-12

Runs alongside the main pathway rather than replacing it, typically as a 12-16 week course timed to a specific test date. Students preparing for both examinations should take the diagnostic for each and commit to the format that suits them.

COMPETITION

Competition Mathematics Levels I-III, alongside any pathway

Placed by assessment rather than by age. Level I aligns to AMC 8, Level II to AMC 10/12, and Level III to AIME and olympiad work. Runs in parallel with the school pathway throughout.

PLACEMENT PRINCIPLE

A pathway is a plan, not a verdict. Վարժարանը does not place a Grade 5 student on a track that determines whether they will see calculus. Acceleration is offered when the evidence supports it and withdrawn when it does not, and a student who moves from accelerated to standard has lost nothing but time pressure. The most common error in American mathematics placement is accelerating a student who is fast but not secure; the resulting gaps surface two years later, in Algebra II, when they are expensive to close.

VERTICAL STRANDS

Ուղղահայաց թեմատիկ գծեր

Five ideas run the length of the curriculum. Teachers are asked to name the connection aloud when a strand resurfaces — a student who is told that the Grade 3 area model is the same picture as the Algebra I binomial product retains both better than a student left to notice it.

NUMBER AND OPERATIONS

Counting and cardinality (G1) → place value and regrouping (G2–G4) → fractions as numbers (G3) → fraction and decimal operations (G4–G5) → ratio and rate (G6) → rational numbers with negatives (G6–G7) → the real numbers and irrationals (G8) → complex numbers (Algebra II).

ALGEBRAIC THINKING

Equal groups and arrays (G2–G3) → the equals sign as balance (G1–G6) → expressions and one-step equations (G6) → multi-step equations (G7) → linear functions and systems (G8) → the function families (Algebra I, Algebra II) → structural analysis and modelling (Precalculus).

GEOMETRY AND MEASUREMENT

Shapes and fair shares (G1) → length and area (G2–G3) → angles and classification (G4) → volume and the coordinate plane (G5) → polygon area and nets (G6) → circles and scale (G7) → transformations, congruence, and the Pythagorean theorem (G8) → formal proof and trigonometry (Geometry).

DATA, STATISTICS, AND PROBABILITY

Picture and bar graphs (G2) → scaled graphs and line plots (G3–G5) → distributions and variability (G6) → sampling and probability models (G7) → bivariate data and the line of best fit (G8) → regression and correlation (Algebra I) → full inference (Statistics, AP Statistics).

REASONING AND PROOF

Explaining a strategy aloud (G1–G3) → justifying with a model (G4–G5) → arguing from structure (G6–G8) → justifying algebraic steps by property (Algebra I) → formal deductive proof (Geometry) → mathematical induction (Precalculus) → written justification under examination conditions (AP courses) → complete olympiad write-ups (Competition).

HIGH-CONSEQUENCE POINTS IN THE CURRICULUM

Ծրագրի առանցքային կետերը

A small number of moments in a mathematics education account for a disproportionate share of later success and failure. These are the points where Վարժարանը deliberately spends more time than the calendar suggests, and where the placement assessment looks hardest.

POINT	WHY IT MATTERS AND WHAT ՎԱՐԺԱՐԱՆ DOES
Grade 3 — Fractions as numbers	Fraction failure in Grades 5–7 is almost always traceable to a Grade 3 unit taught with area models alone. Վարժարանը requires the number line in every fraction lesson. A standalone Fractions Intensive is offered as remediation for students in Grades 4–6.
Grade 3 — Multiplication fact automaticity	The strongest single predictor of Grade 5–7 performance. Fluency work continues in every lesson from Unit 2 of Grade 3 through the end of Grade 4, regardless of the current topic.
Grade 6 — Ratio reasoning	The strongest predictor of Algebra I performance. Cross-multiplication is deliberately withheld until ratio tables, tape diagrams, and double number lines are secure, because students who acquire the shortcut early cannot solve non-routine problems in Grade 7.
Grade 8 — The function concept	Function notation is introduced deliberately early. Students comfortable with $f(x)$ in Grade 8 have a measurably easier Algebra I, and the cost of introducing it early is low.
Algebra II / Precalculus — Algebraic automaticity	Students fail Calculus for algebraic reasons far more often than conceptual ones. Precalculus is treated as an algebra course with a calculus horizon, and cumulative algebra maintenance appears in every assignment.
AP courses — Written justification	The largest source of avoidable AP score loss is insufficient justification, not conceptual weakness. Justification is trained from Unit 1 of every AP course rather than in a spring review.

ASSESSMENT ARCHITECTURE

Գնահատման կառուցվածք

ASSESSMENT	WHEN	PURPOSE
Placement Assessment	Before enrollment	Produces the structured academic profile: current knowledge, gaps, curriculum alignment, and the recommended course, pathway, and pacing track.
Unit Check	End of every unit	Measures the mastery criteria stated in that unit. A failed check returns the student to targeted lessons rather than advancing the syllabus.
Fluency Probe	Every 2nd or 3rd lesson	Short timed retrieval of the skills the current course assumes. Prevents the silent decay that causes mid-year decline.

ASSESSMENT	WHEN	PURPOSE
Diagnostic Checkpoint	At high-consequence points	Fractions in Grades 3–5, ratio reasoning in Grade 6, the function concept in Grade 8, algebraic automaticity before Calculus. Failure triggers remediation before new content is layered on.
Mid-Year Assessment	Roughly halfway	Cumulative. Triggers a pacing review and, where warranted, a pathway change.
End-of-Course Assessment	End of course	All course outcomes. Determines the next course, the pathway, and the pacing, and produces the formal readiness recommendation reported to the parent.

Վարժարան reports **mastery, not grades**. Every topic in a student's profile is marked Mastered, Developing, or Needs Review against the criteria stated in the relevant unit of the relevant syllabus. A topic is marked Mastered only when the criterion is met on two separate occasions at least a week apart, which prevents short-term recall from being recorded as durable learning. Topics marked Needs Review re-enter homework and the opening minutes of subsequent lessons automatically until they are re-mastered.

WHAT EVERY SYLLABUS CONTAINS

Ինչ է պարունակում յուրաքանչյուր ծրագիր

Each of the twenty-two course syllabi follows the same structure, so that a teacher moving between courses and a parent reading across years both know where to look:

- 1 Course description and the Վարժարան teaching approach for that level.
- 2 Placement and readiness indicators, with what each tells the teacher.
- 3 Course learning outcomes.
- 4 Unit-by-unit breakdown: lesson-level topics, learning objectives, observable mastery criteria, the misconceptions the unit reliably produces with the recommended teacher response, and a teaching note.
- 5 Assessment and mastery policy.
- 6 Pacing options, from intensive remediation to acceleration.
- 7 Homework and independent practice expectations.
- 8 Progress milestones reported to parents.
- 9 Materials and resources.

WHY THE MISCONCEPTION TABLES EXIST

The misconception tables are the part of these syllabi that a teacher will use most and that a conventional syllabus omits entirely. They exist because the single most valuable thing an experienced teacher knows is not what to teach but what will go wrong — and because a one-to-one lesson is the setting in which that knowledge produces the most value. A teacher who recognizes an error as the additive-ratio misconception rather than as carelessness fixes it in one lesson. A teacher who does not reteaches the procedure, and the error returns.

College Board and ACT information in August 2026 and should be re-verified each academic year.